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Student Number

2024
HSC TRIAL EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using a black pen.
- Write your Student Number at the top of every page.

Total Marks – 100

Section I – Pages 3 – 5

10 marks

Allow about 15 minutes for this section.

Sections II-V – Pages 7 – 40

90 marks

- Attempt questions from questions 11 – 30.
- Answer all questions in the booklets provided.
- Allow about 2 hours and 45 minutes for this section.

This paper MUST NOT be removed from the examination room.

Section I

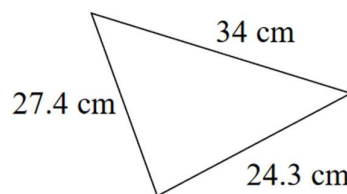
10 marks.

Attempt Questions 1–10.

Allow about 15 minutes for this section.

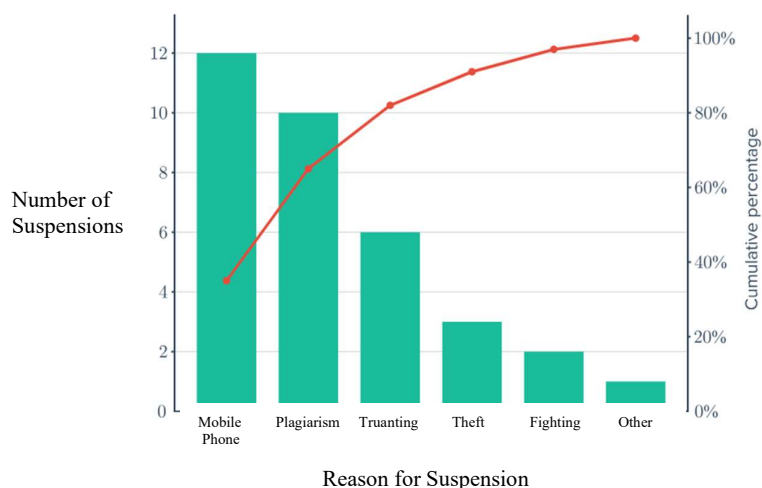
Use the multiple-choice answer sheet for questions 1 – 10.

1. Consider the function $f(x) = |x + 2| + 1$. Which of the following is a true statement?
 - (A) The function is continuous and differentiable for every value of x .
 - (B) The function is not continuous but differentiable for every value of x .
 - (C) The function is not continuous and not differentiable for every value of x .
 - (D) The function is continuous but not differentiable for every value of x .
2. Consider the function $f(x) = \ln(x + \sqrt{x^2 + 1})$. Which of the following describes $f(x)$?
 - (A) An even many-to-one function.
 - (B) An odd many-to-one function.
 - (C) An even one-to-one function.
 - (D) An odd one-to-one function.
3. For what values of k does the quadratic equation $x^2 + kx - 2k = 0$ have real roots?
 - (A) $k \in (-\infty, -8) \cup (0, \infty)$
 - (B) $k \in (-\infty, -8] \cup [0, \infty)$
 - (C) $k \in (-8, 0)$
 - (D) $k \in [-8, 0]$
4. Which of the following gives the magnitude of the smallest angle in the triangle below to the nearest degree?



- (A) 30°
- (B) 45°
- (C) 53°
- (D) 79°

5. The following pareto chart shows the reasons for suspensions at a school.



Which of the following accounted for more than 80% of suspensions?

- (A) Plagiarism.
- (B) Mobile Phone.
- (C) Mobile Phone, Plagiarism and Truancy.
- (D) Truancy, Theft, Fighting and Other.

6. A and B are events such that:

$$P(A \cap B) = \frac{2}{5} \quad \text{and} \quad P(A \cap B') = \frac{3}{7}$$

Which of the following is the value of $P(B'|A)$?

- (A) $\frac{6}{35}$
- (B) $\frac{15}{29}$
- (C) $\frac{14}{35}$
- (D) $\frac{29}{35}$

7. Consider the discrete probability distribution function with random variable X defined in the table below:

x	-2	0	b	$2b$	4
$P(X = x)$	a	b	b	$2b$	0.2

Which of the following is true for the expected value $E(X)$?

- (A) $-0.8 \leq E(X) \leq 1$
- (B) $0 \leq E(X) \leq 1$
- (C) $0.2 \leq E(X) \leq 0.8$
- (D) $0 \leq E(X) \leq 2.4$

8. Which of the following is the value of the gradient of the tangent to $y = -\frac{1}{x}$ at the point $\left(2, \frac{-1}{2}\right)$?

- (A) 4
- (B) -4
- (C) $\frac{1}{4}$
- (D) $-\frac{1}{4}$

9. The graph of a function f is obtained when the graph of function g with the rule $g(x) = x^2 - 3x - 10$ is translated by 7 units in the vertical direction followed by a reflection across the y -axis.

Which of the following is the rule for the function f ?

- (A) $x^2 + 3x - 3$
- (B) $-x^2 + 3x + 3$
- (C) $x^2 - 2x - 22$
- (D) $-x^2 + 2x + 22$

10. If $\int_2^6 f(x)dx = 8$, to which of the following is $\int_4^8 2f(x - 2)dx$ equal?

- (A) 4
- (B) 6
- (C) 16
- (D) None of the above.

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Student Number

Question 11 (4 marks)

Marks

Katie has deliberately designed a biased six-sided die with the following probability distribution for X , the number showing on the uppermost face when the die is rolled.

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1 - \theta}{6}$	$\frac{1 - \theta}{6}$	$\frac{1 - \theta}{6}$	$\frac{1 + \theta}{6}$	$\frac{1 + \theta}{6}$	$\frac{1 + \theta}{6}$

- (a) What values of θ allow for $P(X)$ to be a probability distribution function? **2**

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- (b) Find $P(1 \leq X \leq 4)$ in terms of θ . **1**

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- (c) Find the probability of rolling an even number in terms of θ . **1**

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Student Number

Question 12 (5 marks)

Marks

Janet and Alan are playing a tennis match. The probability of Janet winning the first set is 0.3. After that, Janet's probability of winning a set is 0.6 if she won the previous set, or 0.4 if she lost the previous set. The match will continue until either Janet or Alan wins two sets.

(a) Draw a probability tree diagram to describe the match. 2

(b) Find the expected value of the number of sets that the match will last. 2

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(c) Find the probability that Alan wins, given the match lasted 3 sets. 1

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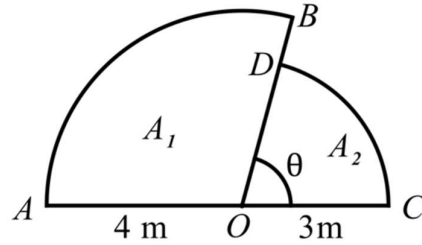
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Student Number

Question 13 (2 marks)

Marks

Two sectors A_1 and A_2 are constructed along the interval AC . $AO = 4$ and $OC = 3$ as shown in the diagram below.



If the area of A_1 is twice the area of A_2 , find the value of θ .

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Question 14 (2 marks)

Prove: $\frac{\cot x + \operatorname{cosec} x}{\tan x + \sin x} = \cot x \operatorname{cosec} x$

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Student Number

Question 15 (5 marks)

Marks

Solve $2 \cos^2 x - 3 \cos x = 2$ for $0 \leq x \leq 2\pi$, leaving your answer in exact form.

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(b) Solve $3 \sin x + \sqrt{3} \cos x = 0$ over the domain $0 \leq x \leq 2\pi$.

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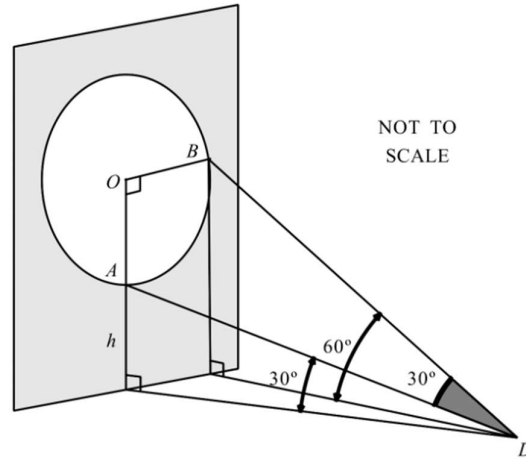
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Student Number

Question 16 (5 marks)

Marks

A circle with centre O and radius 10m is drawn on a vertical wall, standing on a horizontal floor. Point A is vertically below O and $\angle AOB = 90^\circ$ as shown in the diagram below.



A laser pointer is aimed from a point D on the ground. The angles of elevation of A and B from D are 30° and 60° respectively. Let A be h metres above the ground and $\angle ADB = 30^\circ$.
 (a) Write expressions for AD and BD in terms of h . 2

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(b) Hence find the exact value of h . 3

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Student Number

Mathematics Advanced Section III Answer Booklet

21 marks.
Attempt Questions 17–20.

Instructions

- Write your student number at the top of each page.
 - Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your response should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Student Number

Question 17 (7 marks)

Marks

Consider the curve $y = x^3 - 6x^2 + 7$.

- (a) Find the coordinates of any stationary points and determine their nature. **3**

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- (b) Find the coordinates of any points of inflexion. **2**

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Student Number

(c) Sketch the graph, clearly showing all the points found in (a) and (b) and y -intercepts. You do NOT need to find x -intercepts. **2**

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Student Number

Question 18 (5 marks)

Marks

(a) Differentiate xe^{5x} with respect to x .

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(b) Hence evaluate $\int_0^{10} 5xe^{5x} dx$

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Student Number

Question 20 (6 marks)

Marks

The velocity of a particle travelling on the x axis is given by $v(t) = \frac{1}{t+1} - \frac{1}{2}$ for $t \geq 0$.

(a) Show that the particle is stationary when $t = 1$. 1

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(b) Explain why the particle is moving to the left for all times $t > 1$. 2

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(c) Find the distance travelled by the particle in the first 3 seconds, correct to one decimal place. 3

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Student Number

Mathematics Advanced Section IV Answer Booklet

23 marks.
Attempt Questions 21–25.

Instructions

- Write your student number at the top of each page.
 - Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your response should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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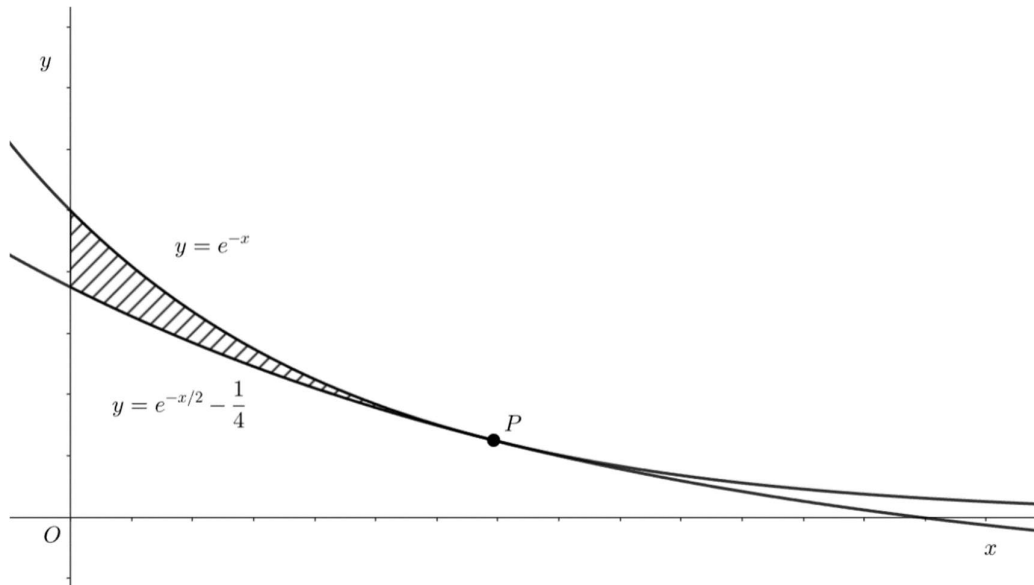
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Student Number

Question 21 (3 marks)

Marks

The curves $y = e^{-x}$ and $y = e^{-\frac{x}{2}} - \frac{1}{4}$ intersect at the point P as shown in the diagram below.



- (a) Show that the point of intersection, P , is $(\ln 4, \frac{1}{4})$.

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- (b) Hence find the area bounded by the two curves and the y-axis, as shaded in the diagram.

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Student Number

Question 22 (2 marks)

Marks

Find:

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$$\int \frac{1-x}{x^3} dx$$

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Student Number

Question 24 (8 marks)

Marks

Consider $f(x) = \ln [\cos(3x)]$ where $-\frac{\pi}{6} < x < \frac{\pi}{6}$.

(a) Find $f'(x)$ in simplest form.

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(b) Show that the equation of the tangent to $y = \tan 3x$ at $x = \frac{\pi}{12}$ is given by

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$$y = 6x + 1 - \frac{\pi}{2}.$$

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Student Number

(c) Find the exact value of the x intercept of the tangent found in (b). 1

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(d) Find the area bound by the curve $y = \tan 3x$, the tangent and the x axis. 3

Give your answer correct to four decimal places.

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Student Number

Mathematics Advanced Section V Answer Booklet

23 marks.
Attempt Questions 26–30.

Instructions

- Write your student number at the top of each page.
 - Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your response should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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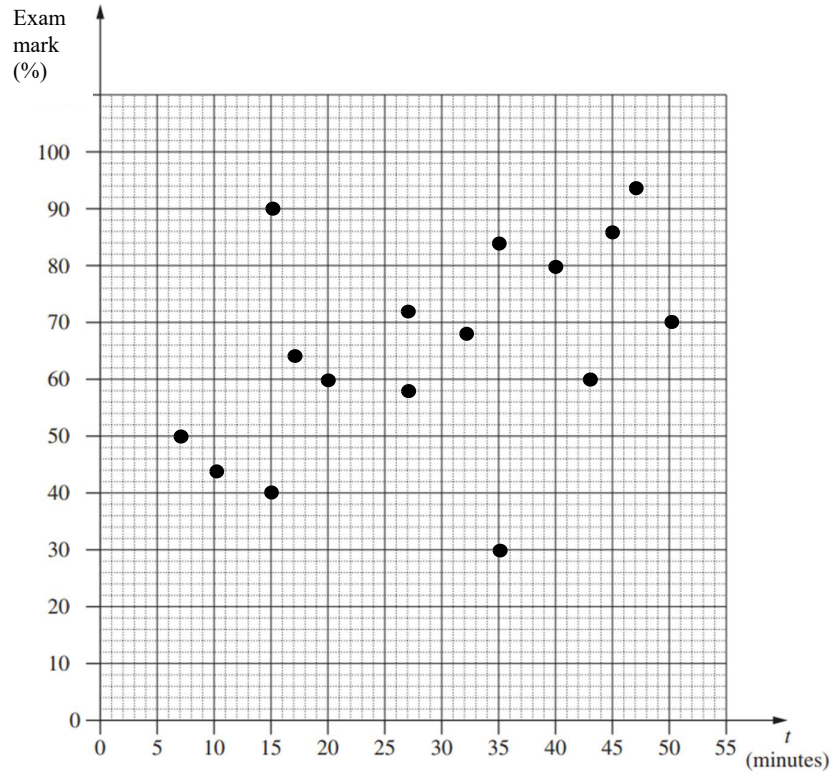
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Student Number

Question 26 (5 marks)

Marks

A researcher conducted a survey of 16 students at a school, asking them for the number of minutes they spent on their device each night and the mark they received in their last examination as a percentage. He plotted the data on the graph below:



(a) Calculate the least squares regression line for the data.

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(b) With evidence, describe the strength of the correlation between minutes spent on a device by a student and their mark in an examination.

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Student Number

(c) A new student is surveyed who spends 20 minutes on their device each night. 1

Predict their score on the next examination to the nearest percentage.

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(d) Another new student wanted to use the model in (a) to predict their next examination 2

but was told that their time spent on their device was too large. What is the smallest amount of time the student could have reported spending on their device? Assume the graduation in time measurement is one minute.

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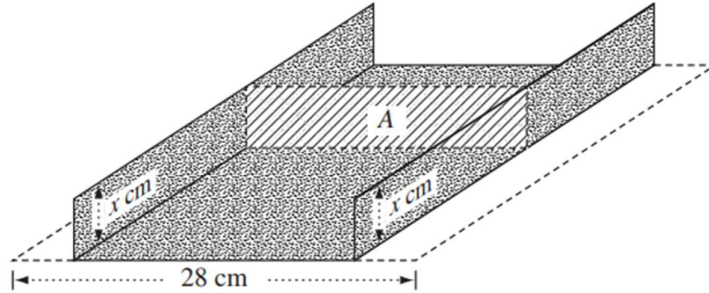
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Student Number

Question 27 (5 marks)

Marks

A long rectangular sheet of metal 28cm wide is to be made into a gutter by turning up sides of equal height x cm, perpendicular to the base as shown below:



(a) Show that the formula for the cross-sectional area, A , of the gutter is

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$$A = 28x - 2x^2$$

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(b) Explain why this formula is only valid for $x \in (0,14)$.

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Student Number

(c) Find the maximum value of A .

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Student Number

Question 28 (4 marks)

Marks

A new artist releases a song on a music streaming platform. The number of ‘listens’ each hour for the first 5 hours is recorded in the table below.

Hour (H)	1	2	3	4	5
Listens (L)	13	36	62	94	138

(a) Show that the predicted number of listens in the 6th hour is 206. 2

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(b) How many predicted ‘listens’ will the song have had at the conclusion of one day? 2

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Student Number

Question 29 (5 marks)

Marks

A toxic substance in a lake is degrading naturally according to the formula $A(t) = 3000m^{-0.2t}$, where A is the number of grams of the substance remaining at t years and $m > 0$ is a constant.

- (a) If $A = 2400$ after 6 years, find the value of m correct to 1 decimal place. **1**

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- (b) Fish can be reintroduced to the lake once the rate of change of the toxic substance is greater than -20 . After how many years can fish be reintroduced? Answer to one decimal place. **3**

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- (c) People can swim in the lake once the amount of the toxic substance is less than 50. In how many years can this happen, to the nearest year? **1**

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